

module-05-dot-addition - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 5 — Bindu Paddhati — Dot Addition (Darśana)

Sanskrit: *Bindu Paddhati* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

A 20th-century mental-arithmetic technique for column addition.

Darśana note. This workshop condenses Module 5's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Show the column $9 + 8 + 7 + 6 = ?$. Most students will sum top-to-bottom. “What if there’s a faster way? A 20th-century Indian arithmetic technique called *Bindu Paddhati* (dot method) gives one. Today we learn it.”

Core idea · 15 min

Bindu Paddhati for column addition.

For each column (units, tens, hundreds): 1. Add running total downwards. 2. Whenever you reach 10 or more, mark a DOT (•) next to that digit and continue with the remainder. 3. At the end of the column, write the remainder. Count the dots — that’s your carry.

Example: $9 + 8 + 6 + 4 = ?$ - $9 + 8 = 17 \rightarrow$ mark • by 8, remainder 7 - $7 + 6 = 13 \rightarrow$ mark • by 6, remainder 3 - $3 + 4 = 7 \rightarrow$ final - Total: 2 dots + 7 = 27. ✓

Speed: much faster than mental running-total for long columns.

Demonstration · 10 min

Live calculation race. Same column ($47 + 38 + 26 + 91 + 53$), two methods: - Traditional: ~25 seconds for most students - Bindu Paddhati: with practice, ~12-15 seconds

Class observes the time difference.

Source moment · 5 min

Paraphrase Tirthaji 1965: “*The Bindu Paddhati removes the need to carry, by using a dot as a visual marker of 10.*” Modern: dot-counting in computer arithmetic uses the same principle (binary carry-saving adders).

Hands-on activity · 15 min

Method race. Each student does 10 column-addition problems using Bindu Paddhati. Time themselves. Then do another 10 using traditional method. Compare times — celebrate any improvement.

Reflection + take-home · 10 min

“*You now know two methods for column addition. When would you use which? Why?*”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Why mental arithmetic? The setup
- **Day 02** — Single-digit + single-digit with the dot
- **Day 03** — Two-digit column addition
- **Day 04** — Three- and four-digit columns
- **Day 05** — Speed practice — Bindu Paddhati vs traditional
- **Day 06** — Why it works — place value + carry
- **Day 07** — Honest history — Tirthaji 1965, Dani 1993 critique
- **Day 08** — Bridge to computer arithmetic — carry-save addition
- **Day 09** — Mixed problem set
- **Day 10** — Project — design your own mental-math poster

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.